Using Graph Deep Learning to Trace Indirect Sourcing in Agricultural Supply Chains

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Abstract

Corporate greenwashing is one of the most problematic areas in sustainable finance research, and is a well-known activity when companies position themselves as sustainable, yet fail to avoid reputational controversies suggesting otherwise. Such disputes can

result from incomplete or conflicting disclosure reports, investigative journalistic inquiries by NGOs or internal corporate communication gaps, and hence are increasingly regarded as a *data*-related challenge, which has potential to be resolved with help of AI. One of such problems, known as *indirect sourcing*, is currently making headlines of many news articles as it reveals the massive scale of existing *blind spots* in evaluations of industrial impacts on nature and climate. The sectors of economy which make direct use of the primary commodities, such as heavy industry, pulp and paper and agriculture are under particular scrutiny as aside from the significant emissions budget they also make impacts on ecosystems, either directly adjacent or distant from their immediate regions of activity. Although supply chains are rarely designed to be transparent in the first place, the current best efforts are primarily undermined by the lack of relevant information, especially on the details of upstream supply chain actors. Using the case of beef industry we modelled spatial relationships between meat packing assets (*abattoirs*) and upstream cattle suppliers (*ranchers*), using fractal principles of the transport infrastructure arrangement. Our results demonstrate that AI can differentiate *direct* and *indirect* suppliers in the upstream chain of each individual meatpacking facility with the confidence 96.7%. This asset-level based approach is the first ever to demonstrate which meatpacking facilities are at risk of stranding due to configuration of their upstream sourcing catchments and exposure to indirect (*bling-spot*) spatial procurement shortcuts. Finally, we provide evidence that this classification approach can be easily translated to other industries, geographies and impact analyses, is both scale and distance agnostic and ethically aligned with the business information protection.

1 Introduction

The subject of indirect procurement is an increasingly trending topic in various economic contexts, where it can appear under such names as *alternative sourcing*, *parallel import* or *secondary supplies*[1–5], and is often associated with opportunities and resilience of the macro- and micro-trade flows[6]. In sustainable finance, however, it usually has more careful connotations[7–11]; As supply chains' transparency requires companies to know what is happening upstream and downstream, and to communicate this knowledge to both internal and external stakeholders, any *blind-spot* on the production line could be associated with potential material risks.

Many financial actors acknowledge the need for better supply chain monitoring strategies, and are seeking to mitigate associated risks with help of the new data sources, algorithms and early warning systems[12–16]. In regards to ESG risks, which are becoming more complex and overarching, companies and businesses with compromised reputational profiles are no longer regarded

in isolation as their adverse performance can seriously compromise ESG profiles (and financial performance) of the downstream participants. Cautious of those risks, the biggest investors, like Storebrand ASA[17], and pension fund KLP[18] sending a clear message to the companies that they do not want to invest into companies linked to Amazon fires and deforestation, which can have a varying *ripple* effects for other financiers, knowingly or unknowingly exposed to the climate and environmental risks in their asset portfolios[19]. Added together, these types of alternative information fall into category of 'missing data'[20], which is a well-known problem in the mainstream finance. In case of sustainable finance applications the situation is even more dire; The requirements geared around precise localisation of the economic activities coupled with the need for semantic descriptors (such as ownership change, production output amendments and exact naming of financial entities) make it particularly difficult information for investors to get hold of in the open domain[21].

Spatial finance[22, 23] was therefore conceived as a first major methodology to embark on this issue in order to both fill the environmental finance information gaps as well as to enable fairer access to the globally consistent data covering major polluting economic sectors. The ultimate vision of the spatial finance is to be able to monitor and evaluate on-site economic activities using remotely sensed data and ground sensors (a.k.a., 'primary data'), whilst also enriching these online metrics with semantic information, derived from disclosures, news articles and other sources of publicly available text data.

Data on agricultural supply chains have been an object of the particular interest in the 'green' finance industry[24], particularly due to the industry's complex and obscure sourcing patterns, and growing fragmentation and encroachment onto the major carbon sinks, such as forests and wetlands[25–27]. Tropical deforestation is estimated to cause about one-quarter of anthropogenic carbon emissions, loss of biodiversity, and other environmental services[28–31]. United Nations Framework Convention for Climate Change (UNCCC) talks are now considering such mitigation mechanisms as avoided deforestation and forest restoration (*reforestation*) to be introduced into monitoring of the sourcing practices in order to identify environmental risks across the entire supply chains of the agricultural commodities[32]. Thus, it has been estimated that stopping the conversion of rainforests into pasture for beef production could reduce Brazil's agricultural carbon emissions by 69%, help slow the sixth mass extinction of species and maintain a crucial carbon sink, vital for cooling the climate[25, 33, 34].

A chain of actors from cattle ranchers through to multinational beef traders, international financiers and the governments that regulate them have been repeatedly accused in either destroying the Amazon directly or not doing enough to mitigate the catastrophic speed of forest destruction, closely associated with the human rights abuses against indigenous and landless groups[30, 31, 35–37]. In Brazil, the biggest meatpacking companies source cattle from thousands of independent ranches, which form a network, deeply implicated in tropical deforestation[29, 38]. It has been already recognised

elsewhere that the companies fail to account for deforestation footprints events in their *direct supply* networks, which account for nearly 30% of their deforestation impacts. The other 70% come from so called *indirect ranches*, which sell the livestock to the direct suppliers and thus often remain unaccounted for[39–42]. Investigations from NGOs and other organisations cite several reasons behind emergence and proliferation of the 'secondary' supply networks[39, 43, 44]. The most frequent one is the lack of direct infrastructure connectivity between cattle ranches and logistically close slaughtering facility[28, 42, 45, 46]; Whereas other findings suggest that 'indirect' suppliers tend to be farms much smaller in size emerging ranches on the fringes of isolated settlements and natural areas[39].

Governmental investigations and those ones from the international NGOs in this sphere differ. Thus, national approach for differentiating *direct* and *indirect* suppliers in Brazilian beef procurement chains is based on surveys and online governmental portals, containing animal transit guides ($GTAs$) and ranch parcel boundaries ($CARs$)[39]. This information is usually assembled independently for each region, known to be at risk from diffuse deforestation (e.g., Mato Grosso, Para), and is rarely translated from Portuguese or merged into one harmonised national database. Global NGO groups, such as MightyEarth, Global Witness and Trase then make use of the regional reports, and after coupling them with earth observation records in order to estimate exact areas of illegal deforestation footprints, issue into a public domain[39, 44, 47]. In some instances, such reports provide detailed information on the downstream customers of such commodities, including big corporations, at the level of company and sometimes at the level of the concrete meatpacking facility. The finest spatial resolution of regions under inquiry is usually municipalities level, which is regarded granular enough for attributing environmental harms to international exports. For instance, Trase[1] tracks Brazilian beef exports from the municipality where the cattle was raised, through to 150 importing countries, identifying the companies involved in this trade. For tracing cattle movements before it reaches slaughterhouses, Trase makes it possible to see whether it has come from a municipality where there is a high risk of deforestation; Owing to this information it was possible to estimate that Brazil's cattle exports are associated with 65,000-75,000 hectares of the forest loss per year[42].

Aware of the growth and accumulation of open investigative reports on the sourcing routes by NGOs, we decided to convert them into one consolidated and curated dataset and verify whether we can come up with the nationally scalable method, which would be also generalise across other agricultural commodities. Graph machine learning has already been tested on similar tasks, where feature propagation proved to be an efficient solution for 'percolated' networks, and is gaining increased attention both within computer science

[1] https://www.trase.earth/

community and downstream application domains alike, due to their capacity to work on multiple data modalities and account for various relational biases[48–51].

In order to construct a functional training graph structure, we processed the collection of open reports, published by the MightyEarth[2] and Global Witness organisations, and extracted training data for the total of 610 farms, covering suppliers for JBS[3], Marfrig[4] and Minerva Global Foods[5] companies (covering 80% of the domestic meat market). A separate list of meatpacking facilities has been assembled for each of the three companies, in order to verify whether 'direct' and 'indirect' sourcing strategies have pervasive pattern at the corporate level, or are a mere artefact of the spatial configuration of the transport infrastructure in Brazil (see Section 4 (Methodology)).

2 Results

Our results provided some clear statistical evidence that corporate sourcing strategies indeed tend to conform to the rules of regional transport connectivity; And these claims are also supported by the previous qualitative analyses of the cattle transportation patterns across the vast distances in Brazil and other countries of the Latin America. Majority of existing studies also suggest that quality and connectivity of the transport infrastructure define relationships between assets in the production chain, where in case of the cattle movements in Brazil, roads infrastructure is critical. However, on its own, this information is insufficient to differentiate 'direct' and 'indirect' suppliers. The adopted partial graph training approach, where we used information on the known fraction of abattoirs and their upstream supply patterns allowed us to train representative structure, which, when deployed on the network of the national roads in Brazil, provided classification confidence level of 96.7%. Results also allow to make a simple conclusion on the subject of 'direct'-'indirect' abstraction rules when modelling sourcing patterns of the primary commodities; Thus, municipalities which have regional centres lying directly on the major routes of the upstream transportation catchments of the meatpacking facilities (*abattoirs*) tend to host the direct cattle supply ranches, whilst the municipalities, which are crossed by the same catchment routes, but have centres located elsewhere, tend to be indirect suppliers (Fig 1). These findings have been confirmed by both deep learning validation process and manual validation, performed on the data entries, collected by the Global Witness investigative data analytics study[39].

Validation of the results using farm-level data demonstrated that this method is by far is the most accurate and flexible, as it can handle complex configurations of municipalities outlines, and also opens up routes for disambiguation of the sourcing patterns even in locations, where facilities of

[2] https://www.mightyearth.org/reports/
[3] https://www.dropbox.com/s/xif0nj0hdkd4r0a/JBS%20ranches.7z?dl=0
[4] https://www.dropbox.com/sh/de1sdcg4o8qhyqc/AAAmWrkoDqmPMBBZ-uHJZEP-a?dl=0
[5] https://www.dropbox.com/sh/phnumh7ozgv43zn/AACUt5Z5SqaSsL7aQJxKgVdea?dl=0

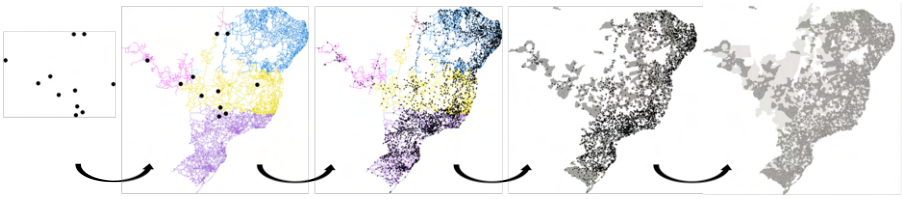

Fig. 1 National sourcing graph model deployment stages (example Minerva SA), starting from the location points of the meatpacking facilities, allocation of road accessibility catchments, definition of the regional nodes and their respective municipalities as 'direct' sourcing regions, and secondary catchments as 'indirect' suppliers

the different companies co-occur (Fig 2). This demonstrates opportunities for AI to perform attribution of environmental impacts by the processing facilities with much higher accuracy than currently being achieved by existing methods, which usually utilise distance radii and other types of Euclidean approximations.

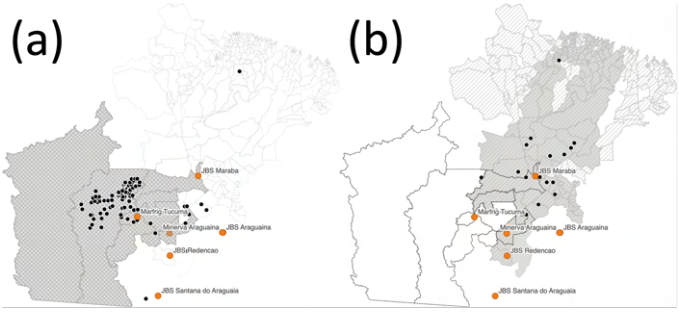

Fig. 2 Excerpt of the region Para, where cluster of abattoirs, belonging to JBS SA, Marfrig and Minerva SA co-occur, and where municipalities are correctly attributed by the model to meatpackers of Marfrig Tucuma (a) and Minerva Araguaina (b).

These results have a number of implications for the quantitative sustainable finance field. First of all, the possibility to disambiguate sourcing regions with the high degree of precision allows to compare companies and their risks of exposure to various physical and climate risks, or quantify their own propensity to cause environmental impacts (such as deforestation, water and air pollution or soil degradation, etc.) (Fig 3)

And, finally, from the regulatory perspective perspective, identification and attribution of 'indirect' supply sourcing regions to the level of processing asset can help companies and investors to locate and quantify the risks assets stranding, based on the concrete spatial evidence. (Fig 4)

3 Sequential graph algorithm

For the neural graph abstraction we propose an implementation of the minimum spanning tree (MST) [52, 53], which is well suited for modelling roads

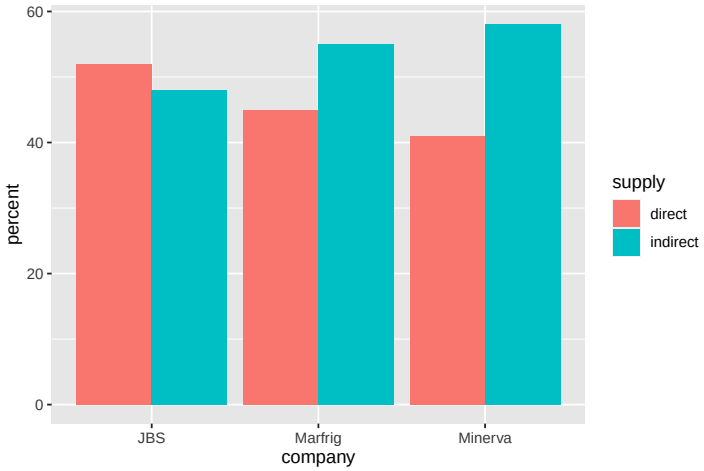

Fig. 3 The shares of potential 'direct' and 'indirect' sourcing regions for 'Big 3' meat producers in Brazil, based on the neural graph simulations of the roads infrastructure and composite situation of their meat-packing facilities within it

networks with embedded differentiated administrative nodes. In order to be able to account for the secondary iteration of MST on the downstream municipality centers, we use graph machine learning[48–50, 54], which inputs training example, covering both MST repetitions (i.e., *fractal MST*) and deploys it on the entire ensemble 'abattoirs-roads-urban centres'.

$$Prim : x_i^{(1)} = \begin{cases} 1 & i = s \\ 0 & i \neq s \end{cases} \quad Prim : x_i^{(t+1)} = \begin{cases} 1 & x_i^{(t)} = 1 \\ 1 & \boxed{\underset{j\ s.t.\ x_j^{(t)}=0}{\operatorname{argmin}} \ \underset{k\ s.t.\ x_k^{(t)}=1}{\min} \ \mathrm{e}_{jk}^{(t)}} \\ 0 & otherwise \end{cases} \tag{1}$$

$$Prim : p_i^{(t)} = \begin{cases} i & i = s \\ p_i^{(t-1)} & i \neq s \wedge x_i^{(t)} = 1 \\ \boxed{\underset{j\ s.t.\ x_j^{(t)}=1}{\operatorname{argmin}} \ e_{ij}^{(t)}} & x_i^{(t)} = 0 \wedge x_i^{(t+1)} = 1 \\ \perp(undefined) & otherwise \end{cases} \tag{2}$$

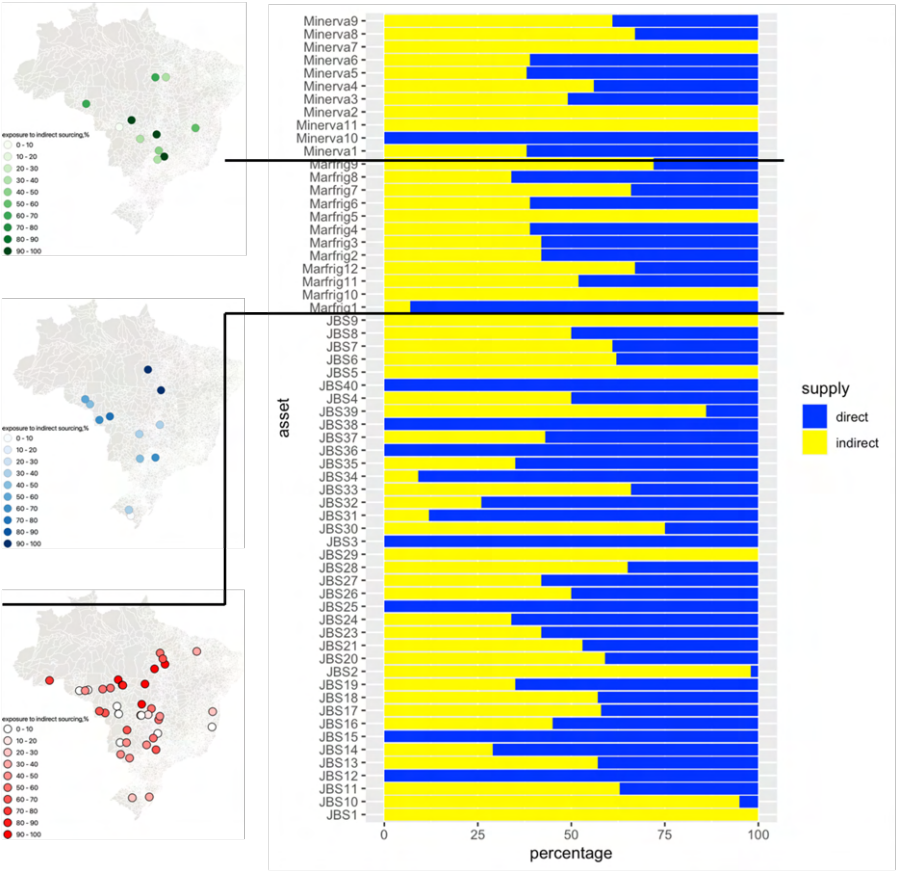

Fig. 4 Visualising exposure proportions to 'indirect' sourcing for each individual meat-packing facility of JBS SA, Marfrig and Minerva SA companies in Brazil. Higher propensity levels illustrate where procurement *blind spots* are likely to occur and mask out potential environmental harms and other types of corporate misconduct.

4 Methods

4.1 Datasets

4.1.1 Investigative reports from *Mighty Earth* and *Global Witness*

We used openly available text reports, published on the websites of the NGOs Mighty Earth[6] and Global Witness[7], which are structured in Portable Documents format (*pdf*). The essential extracted variables of interest were:

- Coordinates of the cattle-rearing ranches, and their names;

[6] https://www.mightyearth.org/reports/
[7] https://www.globalwitness.org/en/campaigns/forests/beef-banks-and-brazilian-amazon/

- Where coordinates weren't available, we used the Environmental Rural Registry (CAR) unique IDs of the farms, which then subsequently were georeferenced;
- Names of the abattoir(s), where each farm supplied cattle to directly;
- Names of the abattoir(s), where each farm supplied cattle to indirectly;
- Names of the transit municipalities (where available)(Fig 5).

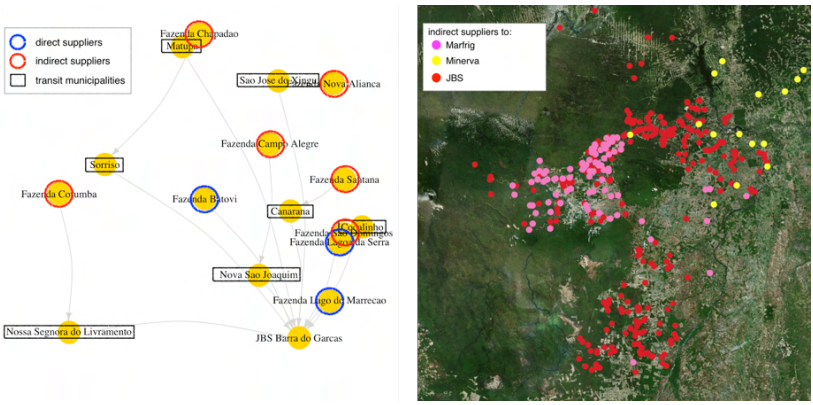

Fig. 5 Examples of the data entries formats, extracted from Mighty Earth *(left)* and Global Witness *(right)*

4.1.2 National roads infrastructure

We used OpenStreetMap roads database, which is free and open global dataset, available for direct download from the Feofabrik.de[8] or directly live using Overpass API[9](Fig 6).

4.1.3 Asset-level data for JBS SA, Marfrig and Minerva SA meatpacking facilities (Brazil)

The complete inventory of meat-packing facilities of JBS SA, Marfrig and Minerva SA has been manually assembled as a regularly curated dataset, which is available for download from [...].

4.2 Methods

We used TensorFlow implementation of the graph learning abstraction[10], in order to initiate Prim-Dijkstra algorithm, accounting for the following prior assumptions:

[8] http://download.geofabrik.de/south-america/brazil.html
[9] http://overpass-api.de/
[10] https://github.com/deepmind/graphnets

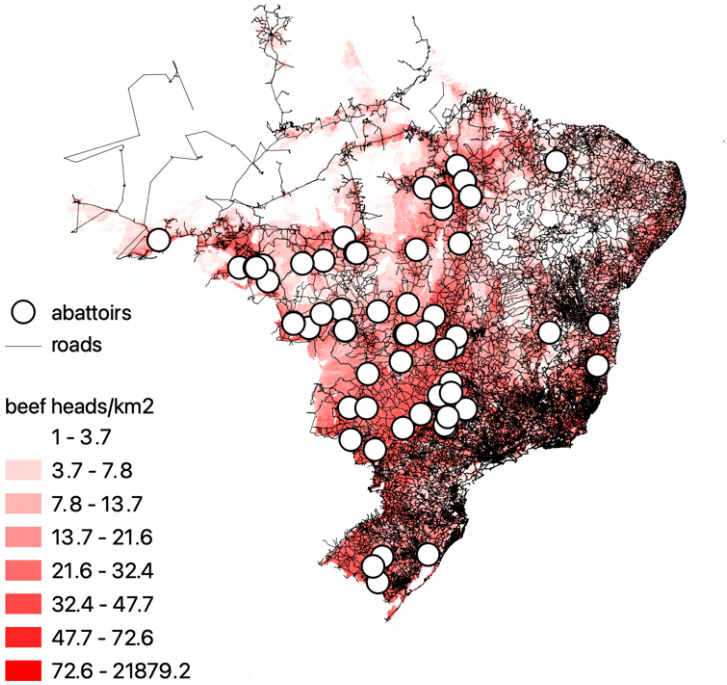

Fig. 6 Network of the primary, secondary and tertiary roads in Brazil, overlaid onto the beef density basemap and associated with the meatpacking facilities of the major beef producing companies

- Cattle can travel from district to district and pass by dozens of farms before it reaches the meat-packing facility;
- Municipalities can specialise as both 'direct' and 'indirect' suppliers, depending on the transportation catchment they belong to;
- Relationships between 'direct' and 'indirect' suppliers aren't defined by the distance thresholds;
- Sourcing municipalities aren't strictly specialised; i.e., some regions can be both 'direct' and 'indirect' suppliers depending on their position in the supply chain hierarchy of the particular company;
- Since all current models are predominantly export-orientated, design of the unified export-domestic model is advantageous, especially for cases where environmental footprints are attributable to domestic consumers (domestic consumption of beef in Brazil account for 80% of the total production).

The training dictionary was based on the georeferenced asset information, extracted from the publicly available NGO reports, and aggregated at the level of municipalities (170 in total), primarily concentrated in the regions Para and Mato Grosso. Complete classifier was subsequently deployed onto entire national network of municipalities (around 5,000), containing only two types of the node labels, indicating their belonging (or exclusion) from the respective for each abattoir MST branch.

5 Conclusion and Discussion

This paper presents novel algorithm, which can classify and attribute 'direct' and 'indirect' suppliers to the individual processing facilities of three biggest beef producers in Brazil. The implications of being able to *spatialise* sourcing catchments with the high degree of confidence is significant for the field of sustainable finance; Hence we should be able to not only profile the companies in terms of their abstract supply chain activities, but also associate the multiplicity of financial and reputational risks companies may be exposed to, but currently unaware of due to existing *blind spots* in the lengthy and complex upstream procurement chains.

Agricultural supply chains are known to lack the transparency, and the situation is particularly problematic where sourcing of the core 'cash' commodities coincides with the *hotspots* of biodiversity and natural carbon stocks, such as tropical forests, wetlands and oceans. The underlying physics of the sourcing is therefore an important piece of information, which could be of interest to sustainable investors and other actors of financial field, seeking to implement sustainable practices into their core business activities.

Graph deep learning algorithm, which have been chosen as a major method of abstraction, demonstrated a remarkable performance, and there is a need to understand how it can also generalise across other countries and cover additional controversial primary commodities, such as soy, cocoa and palm oil.

Supplementary information.

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Declarations

Funding.

Competing interests.

Availability of data and materials.

Code availability.

Authors' contributions.

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